Class 9

Chemistry

Volume 2

Solutions

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5. Ionic Equilibrium

WORKSHEET – 1

Level – I

1) C	2) A	3) D	4) D	5) C	6) B	7) A	
8) C	9) D	10) A					

Level – II

1) A	2) D	3) A	4) D	5) C	6) D	7) B	8) A
9) C	10) B	11) D	12) B	13) A	14) B	15) A	16) C
17) D	18)A,B,C,D		19) A,B,C,D	20) A,B	21) D	22) A	23) B
24) a-p,r;b-s;c-q;d-s			25) a-q,r;b-q,r;c-p,s;d-p,s			26) C	27) D
28) D							

WORKSHEET – 2

Level – I

1) D	2) B	3) B	4) B	5) B	6) A	7) D	8) B

Level – II

1) B	2) C	3) D	4) D	5) A	6) D	7) A	8) B
9) B	10) A	11) B	12) D	13) A	14) C	15) B	16) B
17) A	18) A	19) A	20) A,B,C,D	21) A	22) C	23) A	
24) a-p,q,r,s; b-p,r; c-r; d-p,q,s			25) C	26) D	27) B	28) B	

2. $K_a = 10^{-5}$

$$P^{k_a} = 5$$
$$P^{K_a} + P^{K_b} = 14$$
$$P^{kb} = 14 - 5 = 9$$

9. No.of NaOH $= \frac{4}{40} \times 1 = \frac{1}{10} = 0.1$

N = Concentration of

$[OH^-] = 1 \times 10^{-1}$ mol/ litre

15. $\alpha = 1, K_a = 1.8 \times 10^{-5}$

$K_a = C\alpha^2$

$$C = \frac{K_a}{\alpha^2} = \frac{1.8 \times 10^{-5}}{\frac{1}{100} \times \frac{1}{100}}$$

$= 1.8 \times 10^{-5} \times 10^4$

$= 1.8 \times 10^{-1}$

$\Rightarrow [H^+] = 1.8 \times 10^{-1} = N$

$$N = \frac{x}{60} \times 1$$

$$1.8 \times 10^{-1} = \frac{x}{60}$$

$= 1.8 \times 10^{-1} \times 60$

$= 0.18 \times 60 = 10.8$ g.

16. $\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{K_{a1}}{K_{a2}}}$

$$= \sqrt{\frac{2.1 \times 10^{-4}}{1.1 \times 10^{-5}}} = \frac{1.449 \times 10^{-2}}{1.048 \times 10^{-3}}$$

$= \frac{14.4}{1.038} = 13.8$

$\alpha_1 : \alpha_2 = 13.8 : 1$

17. $N = \frac{0.16}{32} \times \frac{1000}{500} = 10^{-2}$

Concentration (C) $= 10^{-2}$

$\alpha = \sqrt{\frac{K_b}{C}} = \sqrt{\frac{4 \times 10^{-6}}{10^{-2}}} = 2 \times \frac{10^{-3}}{10^{-1}} = \frac{2}{100} = 2\%$

18. $K_a = 1.6 \times 10^{-5}$

$CH_3COOH \rightleftarrows CH_3COO^- + H^+$

C 0 0

$C(1-\alpha)$ $C\alpha$ $C\alpha$

$0.01(1-\alpha)$ 0.01α 0.01α

$0.01(1-\alpha)$ $0.01\alpha(0.01\alpha + 0.1)$ In presence of 0.1MHCl

$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$

$1.6 \times 10^{-5} = \frac{0.01\alpha \times (0.01\alpha + 0.1)}{0.1(1-\alpha)}$

$\alpha = 0.016\%$

19. Concentration (C) of the acid $= \frac{N}{20} = 0.05$ N out of 100 molecules,

The no. of dissociated molecules = 3.5

$\alpha = \frac{3.5}{100} = 0.035$

$K_a = C\alpha^2 = 0.05 \times 0.035 \times 0.035$

$= 6.34 \times 10^{-5}$

21. $H_2O \rightleftharpoons H^+ + OH^-$

Molecules of water dissociated

$= [H^+] = [OH^-] = 10^{-2}$

Total moles of water in 1 L. $= \frac{1000}{18} = 55.55$

Percentage Degree of dissociation of water $= \frac{10^{-7} \times 100}{55.55} = 1.8 \times 10^{-7}\%$

22. $HA \rightleftharpoons H^+ + A^-$

$K_a = \frac{[H^+][A^-]}{[HA]}$

$[H^+] = \sqrt{K_a \cdot C} = \sqrt{2 \times 10^{-5} \times 0.2}$

$= \sqrt{4 \times 10^{-6}} = 2 \times 10^{-3}$

25. $\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{K_{a1}}{K_{a2}}} = \sqrt{\frac{4.5 \times 10^{-4}}{1.8 \times 10^{-5}}}$

$= \sqrt{\frac{4.5 \times 10^{-4}}{1.8 \times 10^{-5}}}$

$$= \sqrt{\frac{45}{18} \times 10}$$
$$= \sqrt{2.5 \times 10}$$
$$= 5$$
$$\alpha_1 : \alpha_2 = 5 : 1$$

26. $\alpha = \sqrt{\frac{K_a}{C}} = \sqrt{\frac{1\times10^{-5}}{0.1}} = \sqrt{10^{-4}} = 10^{-2}$

WORKSHEET – 3

Level – I

1) B	2) A	3) A	4) D	5) B	6) B	7) A	8) B
9) D	10) C						

Level – II

1) B	2) C	3) D	4) B	5) B	6) D	7) C	8) D
9) A	10) B	11) D	12) B	13) D	14) C	15) C,D	16) A
17) B	18) B	19) a-r,s; b-p,s; c-p,s; d-p			20) A,B	21) a-q; b-p; c-r;d-s	
22) B	23) D	24) D	25) a-s;b-p;c-q;d-r,s		26) D	27) a-p;b-q;c-r;d-s	
28) A							

1. $K_w = [H^+][OH^-] = 10^{-6} \times 10^{-6} = 10^{-12}$
   $[\because [H^+] = 10^{-6}$, purewater at 90°C$]$
   $[P^H = 6, P^{OH} = 6]$
2. $K_w = 10^{-14}$
3. $[H^+] = [OH^-]$
   $K_w = [H^+][OH^-]$
   $K_w = [H^+]^2$
   $[H^+] = \sqrt{K_w}$
4. $$N = \frac{N_1\ V_1 + N_2\ V_2}{V_1 + V_2} = \frac{10^{-3}\ V + 10^{-2}\ V}{2\ V}$$
   $$= \frac{V10^{-2}(10^{-1} + 1)}{2\ V}$$
   $$= 1.1 \times 10^{-2} = 0.55 \times 10^{-2}$$
   P^H of $A = 3, [H^+] = 10^{-3} = N_1$
   P^H of $B = 2, [H^+] = 10^{-2} = N_2$
   $P^H = -\log 0.55 + 2\log 10 = 1.74$
5. $[H^+] = 10^{-6}$ at 70°C
   $[OH^-] = 10^{-6}$
   $\because [H^+] = [OH^-]$
6. $P^H = 14 - 3 = 11$
   $[OH^-] = 0.001 = 10^{-3}$
   $P^{OH} = 3$
   $P^H + P^{OH} = 14$
   $P^H = 14 - P^{OH}$

$P^H = 14 - 3 = 11$

P^H of KOH $= 10, P^{OH} = 4$; P^H of

7. NaOH $= 12, P^{OH} = 2$

$[OH^-] = [N] = 10^{-4}$ $[OH^-] = [N] = 10^{-2}$

$N_{NaOH} = XN_{KoH}$ $10^{-2} = X10^{-4}$

$$x = \frac{10^{-2}}{10^{-4}} = 10^{+2}$$

8. $N = \frac{0.4}{40} \times 1 = \frac{1}{100} = 10^{-2}$

$P^{OH} = 2$

$P^H = 12$

$[H^+] = 10^{-12}$

9. $M = \frac{n}{v(lit)}, P^H = 13$

$$10^{-13} = \frac{n}{1 \text{ lit.}}$$

$n = 10^{-13}$

$1000ml.\ 10^{-13}$mols.

$1ml. \longrightarrow ?$

Number of moles in 1ml.

$= \frac{10^{-13}}{1000} = 10^{-16}$ moles

$n = 10^{-16}$ moles

1 mole $\rightarrow 6.023 \times 10^{23}$ ions

10^{-16} moles $\rightarrow 6.023 \times 10^{23} \times 10^{-16}$

$= 6.023 \times 10^7$ ions

11. $P^H = 5$

$[H^+] = 10^{-5}$

Diluted 1000 times it means 10^3

$[H^+]$ions decreases

$$[H^+] = \frac{10^{-5}}{10^3} = 10^{-5} \times 10^{-3} = 10^{-8}$$

$[H^+] = 10^{-8}$

$P^H = 8$. But acids Never have P^H Greater than 7

$\therefore P^H = 6 - 7$

12. $M = \frac{n}{V}, P^H = 6, [N] = 10^{-6}$

$n = 10^{-6} \times 1.3$

13. $\frac{N_1 V_1 + N_2 V_2}{V_1 + V_2} = N_R$

$$\frac{1 \times 0.1 + 10^{-4} \times 1000}{1001} = N_R$$

$N_R = 0.2 \times 10^{-3}$

$P^H = -\log 0.2 \times 10^{-3} = 2.3$

14. $N = \frac{N_A V_A - N_B V_B}{V_A + V_B}$

$$= \frac{50 \times 0.4 - 50 \times 0.2}{100}$$

$= \frac{20-10}{100} = \frac{10}{100} = 0.1 = 10^{-1}$

$P^H = -\log 10^{-1} = 1$

22. $[OH^-] = 0.05M$

$N = 2 \times 0.05 = 0.1$

$P^H = -\log[OH^-] = -\log 10^{-1} = 1$

$P^H = 13$

23. $N = \frac{N_1\ V_1 + N_2\ V_2 + N_3\ V_3}{V_1 + V_2 + V_3}$

$= \frac{10^{-3} \times V + 10^{-4}\ V + 10^{-5}\ V}{3\ V}$

$= \frac{V^{-3}0^{-3}(1 + 10^{-1} + 10^{-2})}{3\ V}$

$= \frac{10^{-3}(1 + 0.1 + 0.01)}{3}$

$N = \frac{10^3(1.11)}{3} = 0.37 \times 10^{-3} = 3.7 \times 10^{-4}$

28.

$$N_R = \frac{N_1\ V_1 + N_2\ V_2}{V_1 + V_2}$$

$$= \frac{75 \times 0.2 + 25 \times 1}{100}$$

$$= \frac{15 + 25}{100} = \frac{40}{100} = 0.4$$

$N_1\ V_1 = N_2\ V_2$

$0.4 \times 100 = N_2 \times 400$

$N_2 = 0.1 = 10^{-1}$

$P^H = -\log 10^{-1} = 1$

WORKSHEET – 4

Level – I

1) A 2) A 3) A 4) B 5) A 6) B 7) C 8) B 9) C

Level – II

1) A 2) A 3) C 4) A 5) C 6) C 7) D

8) B 9) B 10) C 11) B 12) C 13) B 14) A

15) D 16) C 17) D 18) B 19) D 20) A 21) B

22) a-p; b-q; c-r; d-s,t 23) B,D 24) D 25) C 26) B 27) B

28) a-s; b-p; c-q; d-r

8. P^H of the buffer does not change

9. 0.15 mole NH_4OH

0.25 mole ^{NH_4Cl}

$K_b = 1.98 \times 10^{-5}$

$P^{K_b} = +4.703$

$P^H = 14 - P^{K_b} - \log\frac{[\text{ Salt }]}{[\text{ Base }]}$

$P^H = 14 - (+4.703) - \log\frac{[0.25]}{[0.15]}$

$= 14 - 4.703 - 0.22 = 9.077$

10. 50ml. of $0.1M \cdot CH_3COONa$

50ml. of $0.01M. CH_3COOH$

$P^{Ka} = 4.76$

$P^H = 4.76 + \log\frac{0.1}{0.01} = 4.76 + 1 = 5.76$

11. 50ml. of $0.2M \cdot CH_3COOH$

25ml. of $0XM. CH_3COONa$

$P^{K_a} = 4.8$

$P^H = 4.8$

$4.8 = 4.8 + \log\frac{25 \times C}{50 \times 0.2}$

$1 = \log\frac{C}{0.4}$

$\Rightarrow \frac{C}{0.4} = 10^1 \Rightarrow C = 10 \times 0.4$

$C = 4M.$

13. $K_a = 5 \times 10^{-10} P^{Ka} = 9.3$

$P^H = 9$

v =? ?M. KCN

10ml. of 2M. HCN

$9 = 9.3 + \log\frac{V \times 5}{10 \times 2}$

$-0.3 = \log\frac{v}{4}$

$\frac{V}{4} = 10^{-0.3} \Rightarrow \frac{V}{4} = 0.5011$

$V = 0.5011 \times 4$

$V = 2ml$

14. $P^H = P^{Ka} + \log\frac{[\text{ Salt }]}{[\text{ Acid }]}$

[Salt] = [Acid]

$P^H = P^{Ka} = 5$

15. $K_b = 10^{-4}\ P^{K_b} = 4$

[salt] to [base] = ?

$9 = 14 - 4 - \log\frac{[\text{ Salt }]}{[\text{ Base }]}$

$-1 = -\log\frac{[\text{ Salt }]}{[\text{ Base }]}$

$\frac{[\text{ Salt }]}{[\text{ Base }]} = 10 \ \therefore 10:1$

16.

$$\phi = \frac{\text{no. of moles added per litre}}{\text{change in } P^H}$$

$$= \frac{5}{2} = 2.5$$

18. $10MCHCOONa_3$

$1MCH_3COOH$

$P^{K_a} = 4.8$

$$P^H = 4.8 + \log\frac{10}{1} = 5.8$$

24. $P^{Ka} = 4.5 P^{Ka} + P^{K_b} = P^{K_w}$

$P^{OH} =?\ P^{K_b} = 14 - 4.5 = 9.5$

50% dissociation

⇒ [Salt] = [Acid]

$P^{OH} = P^{K_b} = 9.5$

25. 0.15 mole CH_3COOH

$P^{K_a} = 4.8$

0.15 mole CH_3COONa

0.05 mole $NaOH$

$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

$0.15\ 0.05$

$0.15 - 0.05\ 0.05 - 0.05\ 0.05 + 0.15$

$$P^H = 4.8 + \log\frac{0.2}{0.1}$$

$$= 4.8 + 0.3010 = 5.101$$

26. 0.01 mole HCl

100ml. of $0.2M.\ NH_3$

$P^{K_b} = 4.74$

$0.3M$ in NH_4^+

$NH_3 + HCl \rightarrow NH_4Cl$

$NH_3 + HCl \rightarrow NH_4Cl$

$0.2 \quad 0.01 \quad 0$

$0.2 - 0.01\ 0.01 - 0.01\ 0.01 + 0.3$

$0.19 \quad 0 \quad 0.3$

$$P^H = 14 - 4.74 - \log\frac{0.31}{0.19}$$

$$= 14 - 4.74 - 0.212 = 9.048$$

WORKSHEET – 5

Level – I

1) D	2) D	3) B	4) A	5) C	6) B	7) D
8) D	9) B	10) C				

Level – II

1) A	2) D	3) B	4) A	5) B	6) D	7) B
8) A	9) B	10) A	11) B	12) C	13) A	14) A,B
15) C,D	16) A	17) B	18) A	19) a-p; b-q; c-r; d-s		
20) B	21) A	22) C	23) a-s; b-q; c-r; d-p			
24)A,B,C,D	25) A	26) D	27) B	28) a-q,t; b-r,s; c-q,r; d-p		

7. $K_h = 2 \times 10^{-5}$
$K_b = ?$
$A_2\ B_3\ K_b = \frac{10^{-14}}{2\times10^{-5}}$
$= 0.5 \times 10^{-14} \times 10^{+5} = 5 \times 10^{-10}$

9. Fow W.A + W.B hydrolysis does not depend on concentration
$$h = \sqrt{\frac{K_w}{K_a + K_b}}$$

10. $C = 0.003M$. aq. solution of NaOCN
$K_a = 3.33 \times 10^{-4}$
$$h^2 = \frac{K_w}{K_a C}$$
$$h\% = \frac{K_w}{K_a C} \times 100$$
$$= \sqrt{\frac{10^{-14}}{3.33 \times 10^{-4} \times 0.003}} \times 100 = 0.01$$

11. $P^{K_a} = 14 - P^{K_a} = 14 - 1.62 = 12.38$
$$P^H = \frac{1}{2}[P^{K_w} + P^{K_a} + \log C]$$
$$= \frac{1}{2}[14 + 12.88 + 0] = 13.19$$

12. $K_a = 5 \times 10^{-5}$
$h\% = ?$
$C = 0.02M.$
$$h = \sqrt{\frac{K_w}{K_a \cdot C}} = \sqrt{\frac{10^{-14}}{5 \times 10^{-5} \times 0.02}}$$
$$= \sqrt{10^{-14} \times 10^{6}}$$
$h\% = 10^{-4} \times 100 = 10^{-2}$

13. $K_a = 5.6 \times 10^{-6}$
$P^{K_a} = 5.25$
$NaOH + HA \rightleftharpoons NaA + H_2O$
$0.1 \times V\ \ 0.1 \times V\ \ 0\ 0$
$0\ \ 0\ \ 0.1\ V$
$[NaA] = \frac{0.1v}{2v} = 0.05M.$

$[OH^-] = \sqrt{\frac{K_w}{K_a} \times C}$

$\sqrt{\frac{10^{-14} \times 0.05}{5.6 \times 10^{-6}}} = 9.45 \times 10^{-6}$

$P^{OH} = -\log[OH^-]$

$= -\log(9.45 \times 10^{-6}) = 5.025$

$P^H = 14 - P^{OH} = 14 - 5.025 = 8.975.$

16.

$h\% = \sqrt{\frac{K_w}{K_a \cdot C}} \times 100$

$= \sqrt{\frac{10^{-14}}{10^{-5} \times 0.1}} \times 100$

$= 10^{-4} \times 100 = 0.01\%$

18. $K_b = 1.8 \times 10^{-5}$

$K_a \times K_b = K_w$

$K_a = \frac{10^{-14}}{1.8 \times 10^{-5}} = 0.55 \times 10^{-9}$

$= 5.5 \times 10^{-10}$

25.

$K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{4.5 \times 10^{-10}}$

$= 0.222 \times 10^{-4} = 2.2 \times 10^{-5}$

WORKSHEET – 6

Level – I

1) A	2) B	3) C	4) A	5) C	6) C	7) D	8) A	9) B
10)C								

Level – II

1) A	2) B	3) 3	4) A	5) C	6) C	7) D	8) A	9) D
10) D	11) C	12) 4	13) A	14) A	15)B	16) D	17) B	
18) A,D	19) C	20) B	21) A	22) a-p,s; b-p,r,s; c-p,q,s; d-p,q,s				
23) A	24) B	25) C	26) a-q; b-s; c-p; d-t		27) A,B	28) A,D		

12. $K_a = 1 \times 10^{-5}$

$P^{Ka} = 5$

colour changes when $[In^-] = [HIn]$

$P^H = P^{Ka} = 5$

13. $K_b = 1.0 \times 10^{-10}$

$BOH \rightleftharpoons B^+ + OH^-$

$$K_b = \frac{[B^+][OH^-]}{[BOH]}$$

$$1 \times 10^{-10} = \frac{(1 \times 10^{-3})[OH^-]}{(1 \times 10^{-3})}$$

$$[OH^-] = 1 \times 10^{-10}$$

$$[H^+] = \frac{1 \times 10^{-14}}{[OH^-]} = \frac{1 \times 10^{-14}}{1 \times 10^{-10}} = 1 \times 10^{-4}$$

$$P^H = -\log(1 \times 10^{-4})$$

$$P^H = 4$$

14. $P^{k_b} = 4; P^{k_a} = 10$

Collour chnages when $[In^-] = [HIn]$

$\therefore P^H = P^{Ka}$

$P^H = 10$

16. $[In^-] = \frac{1}{3}$

$$[HIn] = 1 + \frac{-1}{3} = \frac{2}{3}$$

$P^H = 6$

$$P^H = P^{K_{In}} + \log\frac{[In^-]}{[HIn]}$$

$$6 = P^{k_{In}} + \log\frac{1 \times 3}{3 \times 2}$$

$P^{K_{In}} = 6 + 0.3 = 6.3$

WORKSHEET – 7

Level – I

1) D 2) A 3) B 4) D 5) A 6) A 7) A 8) B 9) A 10) C

3. $M_x x_y \rightleftharpoons x^{y+} + yX^{x-}$

$K_{SP} = (xs)^x + (ys)^y = x^x \cdot y^y (S)^{x+y}$

4.

$AgCl \rightleftharpoons Ag^+ + Cl^-$

SS

$K_{SP} = S^2$

5.

$Ag_2SO_4 \rightleftharpoons 2Ag^+ + SO_4^{2-}$

S 2 S S

$K_{SP} = (2S)^2(S) = 4S^3$

6. $CaF_2 \rightleftharpoons Ca^{2+} + 2\ F^-$

$K_{SP} = S(2\ S)^2$

$K_{SP} = 4\ S^3$

$$S = \sqrt[3]{\frac{K_{SP}}{4}}$$

7.

$$M_xX_y \rightleftharpoons xM^{y+} + yX^{-x}$$

$$S \; xS \; yS$$

$$K_{SP} = x^x y^y (S)^{x+y}$$

$$S = \left[\frac{K_{SP}}{x^x y^y}\right]^{\frac{1}{x+y}}$$

Level – II

1) B	2) C	3) C	4) A	5) D	6) C	7) D	8) B	9) B
10) C	11) C	12) A	13) B	14) B	15) C	16) C	17) D	
18)A,B,D	19) A	20) B	21) B	22) a-q; b-r; c-s; d-p		23) A,B		
24) a-q; b-p; c-s; d-r			25) A,B,C	26) B	27) A	28) A		

1. $A_2B \rightleftharpoons 2A^+ + B^{2-}$

S 2 S S

$K_{SP} = (2S)^2(S)$

$K_{SP} = 4S^3$

$$S = \sqrt[3]{\frac{K_{SP}}{4}}$$

2. $MmAn \rightleftharpoons mM^{n+} + nA^{m-}$

S ms ns

$K_{SP} = (ms)^m(ns)^n$

$K_{SP} = m^m n^n s^{m+n}$

3. $K_{SP} = 3.2 \times 10^{-11}$

$Ax_2 \rightleftharpoons A^2 + 2x^-$

S S 2 S

$K_{SP} = S(2S)^2$

$$S = \sqrt[3]{\frac{K_{SP}}{4}}$$

$$= \sqrt[3]{\frac{3.2 \times 10^{-11}}{4}} = 2 \times 10^{-4}$$

4. CaC

$CaCO_3 \rightleftharpoons Ca^2 + CO_3^{2-}$

S S S

$K_{SP} = S^2$

$S = \sqrt{K_{SP}} = \sqrt{9.3 \times 10^{-8}}$

$= 0.000304$moles/lit

Solubility in g/L.

$= \text{mole/ litre } x$
molecular weight of subs tance
$= 0.000304 \times 100 = 0.0304\text{gm/ litre}$

6. $Mg_3(PO_4)_2 \rightleftharpoons 3Mg^{2+} + 2PO_4^{3-}$
$K_{SP} = [Mg^{2+}]^3[PO_4^{3-}]^2$

8. Common ion effect

9. Common ion effect

10. $Mx_2 \rightleftharpoons M^{2+} + 2x^-$
$K_{SP} = S(2\ S)^2$
$K_{SP} = S(2\ S)^2$
$S = 10^{-4}$
$[M^{2+}] = 10^{-4}M$

11. $K_{SP} = 1.25 \times 10^{-9}$
$0.01M.\ Na_2SO_4$
$PbSO_4 \rightleftharpoons Pb^{+2} + SO_4^{-2}$
S
$S\ (S + 0.01)$
$K_{SP} = (S)(S + 0.01)$
In this ' S ' is negligible in concentration of SO_4^{2-}
$K_{SP} = S(0.01)$
$S = \dfrac{K_{SP}}{0.01}$
$S = \dfrac{1.25 \times 10^{-9}}{0.01} = 1.25 \times 10^{-7}$
$M(OH)_3 \longrightarrow M^{+3} + 3OH^-$

12. $(S + 0.4)$
$K_{SP} = (0.4)(3\ S)^3$
$K_{SP} = 0.4 \times 27 \times S^3$
$K_{SP} = 10.8 \times S^3$
$S^3 = \dfrac{K_{SP}}{10.8}$
$S = \left(\dfrac{K_{SP}}{10.8}\right)^{\frac{1}{3}}$

13. Common ion effect

14. $Ca_3(PO_4)_2 \rightleftharpoons 3Ca^2 + 2PO_4^{3-}$
$\dfrac{[Ca^{2+}]}{[PO_4^{-2}]} = \dfrac{3}{2}$
$[Ca^{2+}] = \dfrac{3}{2} \times 3.3 \times 10^{-7} = 4.95 \times 10^{-7}$
$K_{SP} = [Ca^{2+}]^3[PO_4^{2-}]^2$
$= (4.95 \times 10^{-7})^3(3.3 \times 10^{-7})^2 = 1.32 \times 10^{-32}$

15. $Ba(OH)_2 P^H = 10$
$P^{OH} = 4$
$[OH^-] = 10^{-4}$
$Ba(OH)_2 \rightleftharpoons Ba^{2+} + 2OH^-$

$$K_{SP} = [Ba^{2+}][OH^-]^2$$

$$[Ba^{+2}] = \frac{5 \times 10^{-13}}{(10^{-4})^2} = 5 \times 10^{-5}$$

16. common ion effect

17. $PbSO_4 \rightleftharpoons Pb^{2+} + SO_4^{2-}$

$$K_{SP} = \frac{[Pb^{2+}][SO_4^{2-}]}{[PbSO_4]} = 1.8 \times 10^{-8}$$

$HSO_4^- \sim H^+ + SO_4^{2-}$

$$K_a = \frac{[H^+][SO_4^{-2}]}{HSO_4^-} = 1.0 \times 10^{-10}$$

$$\frac{K_{SP}}{K_a} = \frac{[Pb^{2+}][HSO_4^-]}{[PbSO_4][H^+]} = 1.8 \times 10^2$$

$PbSO_{4(s)} + H^+(aq) \rightleftharpoons HSO_4^- + Pb^{2+}$

$$K_{Eq} = \frac{[HSO_4^-][Pb^{2+}]}{[PbSO_4^-][H^+]}$$

$K_{Eq} = 1.8 \times 10^2$

19. $Ag_2CrO_4 \rightleftharpoons {}_2\,{}_S\,{}_{Sg} + Cr_4$

$K_{SP} = 4\ S^3$

$4S^3 = 4 \times 10^{-12}$

$S = 10^{-4}$

20. $S = 1.1 \times 10^{-4}$

$PbSO_4 \rightleftharpoons Pb^2 + SO_4^{2-}$

$K_{SP} = S^2$

$= (1.1 \times 10^{-4})^2$

$= 1.21 \times 10^{-8}$

21. $Mg(OH)_2 \rightleftharpoons Mg^{2+} + 2OH^-$

$K_{SP} = [Mg^{2+}][OH^-]^2$

$$[OH^-] = \sqrt{\frac{K_{SP}}{Mg^{2+}}} = \sqrt{\frac{1 \times 10^{-12}}{10^{-2}}} = 10^{-5}$$

$[OH^-] = 10^{-5} P^{OH} = 5$

$P^H = 9$

24.

a) $HCl \longrightarrow H^+ + Cl^-$

-0.1

$AgCl \longrightarrow Ag^+ + Cl^-$

$S(S + 0.1)$

In this neglect the ' S '

$K_{SP} = S.(0.1)$

$$S = \frac{K_{SP}}{0.1}$$

b)

$NaOH \longrightarrow Na^+ + OH^-$

0.1

$Ni(OH)_2 \longrightarrow Ni^{+2} + 2OH^-$

S 2 S + 0.1

In this neglect the ' S '

$K_{SP} = S(0.1)^2$

$S = \frac{K_{SP}}{(0.1)^2}$

C)

$Ni(OH)_2 \longrightarrow Ni^{2+} + 2OH^-$

$(S + 0.1)2\ S$

in $(S + 0.1)$

' S ' is very small so neglect the ' S ' $\therefore K_{SP} = (0.1)(2\ S)^2$

$4S^2 = \frac{K_{SP}}{0.1}$

$S^2 = \frac{K_{SP}}{0.4} \Rightarrow S = \left(\frac{K_{SP}}{0.4}\right)^{\frac{1}{2}}$

d)

$Al(OH)_3 \longrightarrow Al^{+3} + 3OH^-$

S 3 S

$NaOH \longrightarrow Na^+ + OH^-$

0.1

$Al(OH)_3 \longrightarrow Al^{+3} + 3OH^-$

S $(3\ S + 0.1)^3$

$K_{SP} = S \cdot (0.1)^3$

$S = \frac{K_{SP}}{(0.1)^3}$

27. Less the K_{SP} faster is precipitation

28. When ZnS starts precipitating

$[S^{2-}] = \frac{K_{SP}}{[Zn^{2+}]} = \frac{10^{-21}}{0.1} = 10^{-20}$

At this concentration of $[S^{2-}]$, the concentration of Cd^{2+} is

$$[Cd^{2+}] = \frac{K_{SP}}{[S^{2-}]} = \frac{8 \times 10^{-27}}{10^{-20}} = 8 \times 10^{-7}$$

6. General Organic Chemistry Part – I

WORKSHEET – 1

Level – I

1) D 2) C 3) C 4) B 5) C 6) A 7) C

8) B 9) A 10) D

Level – II

1) C 2) D 3) A 4) A 50 C 6) D 7) B

8) D 9) D 10)C 11)A 12)A 13)A 14)D

15) C 16)B 17)A,B,C,D 18) D 19) B 20) C

21) a-r; b-p; c-s; d-q 22) D 23) B 24) C 25) C

26) a-r; b-s; c-q; d-p 27) 2,3,4 28)3

WORKSHEET – 2

Level – I

1) D 2) B 3) C 4) D 5) A 6) D 7) A

8) C 9) D 10) B

Level – II

1) B 2) D 3) B 4) C 5) D 6) C 7) A

8)A 9) D 10)D 11)B 12)A 13)B 14)C

15)C 16)D 17)D 18) D 19) B 20) A 21) A

22) a-r; b-s; c-p; d-q 23)14 24) A,B,C 25) A,B,D 26) C 27) D

28) B 29) A,B,C

WORKSHEET – 3

Level – I

1) B 2) A 3) B 4)C 5)D 6) B 7) D

8) B 9) C 10) A

Level – II

1) D 2) A 3) A 4) C 5) B 6) D 7) A

8) D 9) D 10) A 11) B 12) D 13) A 14) C

15)B,C 16)B 17) A 18) A 19) a-r; b-s; c-q; d-p 20) B

21)A,B,C,D 22) D 23) A 24) D 25)A,B,C 26)A,B,C

WORKSHEET – 4

Level – I

1) C 2) A 3) B 4) C 5) B 6) A 7) B

8) B 9) A 10) A

Level – II

1) A 2) D 3) A 4) A 5) C 6) A 7) B

8) B 9) A 10) A 11) B 12) A 13) A 14) B

15) C 16) A,B,C 17) A 18) B 19) B 20) C

21) a-r, b-s, c-p, d-q 22) B 23) A,B 24) A,B 25) A, B

WORKSHEET – 5

Level – I

1) A 2) B 3) C 4) B 5) B 6) C 7) B

8) A 9) A 10)C

Level – II

1) C 2) D 3) D 4) B 5) D 6) C 7) D

8) B 9) C 10) B 11) C 12) B 13) D 14) D

15) D 16) A,C 17) C 18) B 19) C 20) a-q, b-r, c-p, d-s

21)A,B 22) B 23) A 24) C 25)C,D 26) B 27) B

28) C

WORKSHEET – 6

Level – I

1) A 2) B 3) B 4) C 5) A 6) D 7) B

8) C 9) B 10) C

Level – II

1) B 2) B 3) D 4) C 5) A 6) B 7) D

8) A 9) B 10) A 11) D 12) A 13) 4 14) 1

15) 2 16) 1,2 17) 1,2,3,4 18) 1 19) 1 20) 2 21) 3

22) a-q; b-r; c-s; d-p 23) 1 24) 1 25) 1 26) 2,4

27) 3 28) 1

WORKSHEET – 7

Level – I

1) A 2) C 3) B 4) C 5) B 6) C 7) A

8) B 9) A 10) C

Level – II

1) D 2) A 3) B 4) A 5) A 6) B 7) B

8) B 9) A 10) B 11) C 12) D 13) C 14) B

15) C 16) B 17) A 18) A 19) B 20) A,B,C,D 21) D

22) A 23) B 24) a-s; b-p; c-t; d-r 25) A,B

26) a-s; b-r; c-q; d-p 27) A 28) A 29) B

7. Isomerism

WORKSHEET – 1

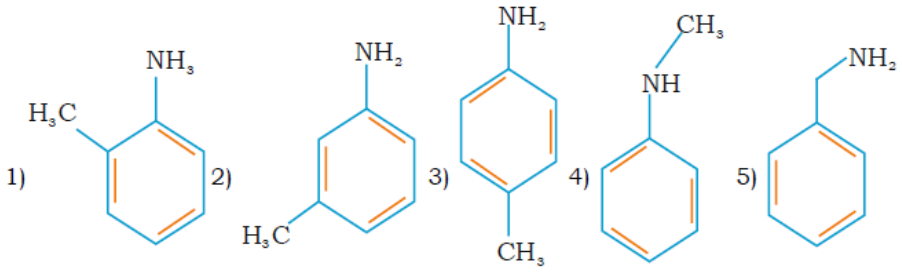
Level – I

1) A	4) B	7) A	9) C
2) A	5) C	8) B	10) A
3) A	6) A		

Level – II

1. A	2. A	3. B	4. D	5. B	6. C
7. B	8. A	9. A	10. A	11. B	12. D
13. A	14. D	15. D	16. B	17. C	18. D
19. A	20. A	21. B, D	22. A	23. a-q, b-r, c-p, d-s	
24. B	25. A	26. D	27. B, C	28. D	29. D
30. C	31. a-q, b-s, c-p, d-r		32. A, B, C		

10.

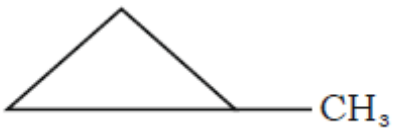


1. $CH_2 = CH - CH_2 - CH_3$

2. $CH_3 - CH = CH - CH_3$

3. $CH_2 = \underset{\displaystyle CH_3}{\overset{|}{C}} - CH_3$

4. 

5. 

WORKSHEET – 2

Level – I

1) B	4) B	7) A	9) D
2) A	5) A	8) B	10) A
3) D	6) D		

Level – II

1. D	5. B	9. D	13. B,C
2. D	6. D	10. D	14. B
3. D	7. D	11. D	15. A
4. A	8. A	12. D	16. B

17. a-p, b-r, c-s, d-p
18. B
19. A, B, C
20. a-q, b-p, c-r, d-s
21. A
22. A
23. B
24. A
25. A
26. C
27. A
28. a-q, b-p, c-s, d-r

WORKSHEET – 3

Level – I

1. D	3. B	5. B	7. C
2. C	4. C	6. C	8. C

Level – II

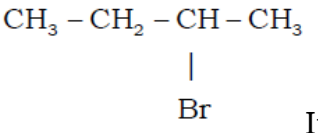
1)A	2) D	3)B	4) B	5)A	6) D	7) B
8) B	9)B	10)A	11)C	12) C	13)B	14)C 15)D
	16)C	17) C,D	18)B	19)B	20) A	21)A,D
22) B	23) C	24) A	25) C			

WORKSHEET – 4

Level – I

1) B	2) B	3) C	4) C	5) C	6) B	7) C
8) C	9) C					

1. The compound which rotates PPL in clockwise direction is called "Dextro rotatory compound".
2. *d* - form is represented by " + "



It contains asymmetric carbon atom

3. Non - super imposable mirror image are called "enantiomers"
4. Meso isomers contains similar organic groups on a symmetric carbons.
5. IInternal compensation of similar groups (−H, 0H) at two asymmetric carbons in tartaric acid
6. Similar groups are attached to the central atoms
7. Resolution is a technique which is used to separate enantiomers from racemic mixture.
8. Dextro rotator is represented by letter " d ".
9. Resolution is technique which is used to separate enantiomers from racemic mixture

Level – II

1) D	2) A	3) D	4) C	5)D	6)A	7) B 8)
A 9) C	10)A	11)A	12)C	13)D	14)C,D	15)A
16)C	17)D	18)a-s; b-r; c-q; d-p		19)A,B,D	20)D	21)C
22)C	23)A,B	24)a-r; b-r; c-p; d-s				

WORKSHEET – 5

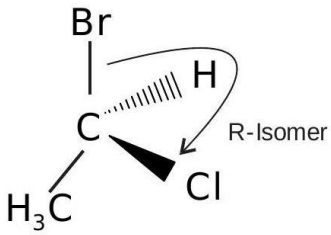
Level – I

1) B	4) C	7) C	10)D
2) A	5) A	8) B	11)B
3) D	6) A	9) B	

Level – II

1) C	2)D	3)C	4)A	5)B	6)D	7) C
8) C	9)C	10)B	11)A	12)B	13)A	14)A
15)D	16)A	17)C	18)C	19)B,D	20)D	21)A
22)B	23)D	24)B	25)C	26)C		

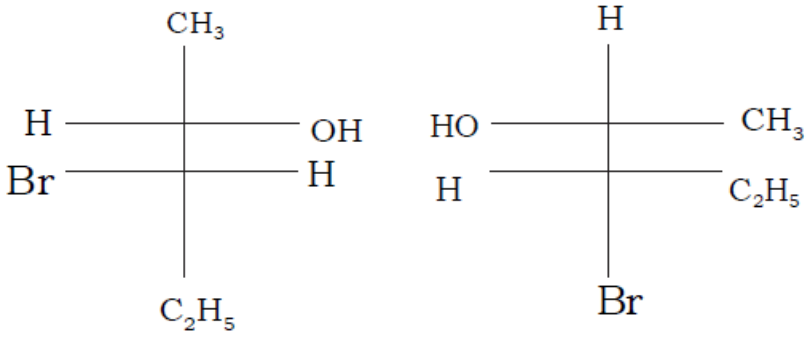
1. Because its dextro rotating −0H on right side
2. Enantiomers are non-super imposable mirror images.



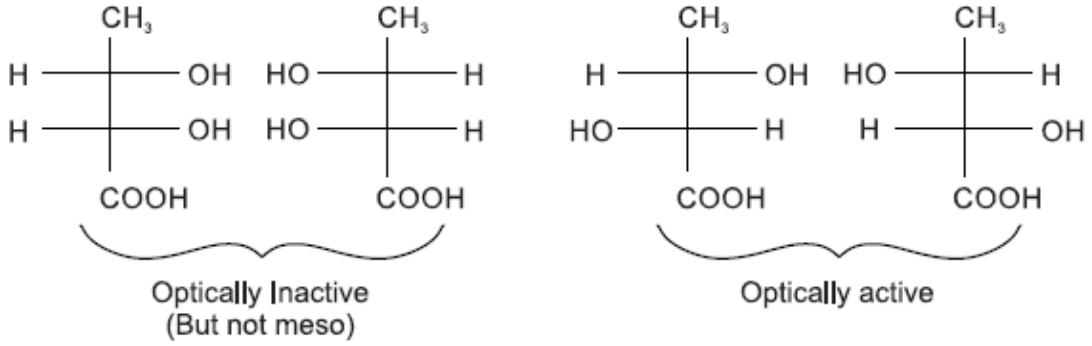
4. Meso compounds are internally compensated form whereas racemic mixtures are externally compensated modification.
5. $CH_3 - \overset{*}{C}H[OH] - CH_2 - CH_3$ caontains one chiral carbon centre. So it gives two enantiomeric forms

isomers = $2^{n-1}(n$ = chiral centres)

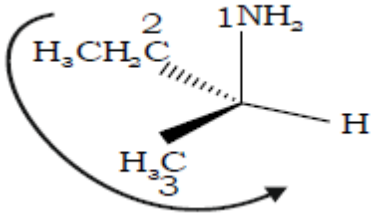
7. Non-super imposable non-mirror images are called "Disstereomers"



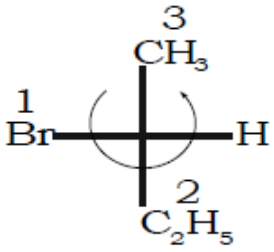
8. Compound are chiral compounds and are having same groups on chiral centres.
9. Meso form is optically inactive form
10. According to R, S nomenclature rules, we should give priority orders for all the groups.
    (i) If (1) → (2) → (3) Priority orders ⇒ Clockwise ⇒ *R*
    Anti clockwise ⇒ S
11. Not mirror images and cis-trans isomers are diastereomers.
12. Internal symmetry not possible. So no meso form



13. Cis-trans isomers are diastereomers
14. Above I,II compounds are non-super impossible and non-mirror images so enantiomers and configurational isomers.
16.

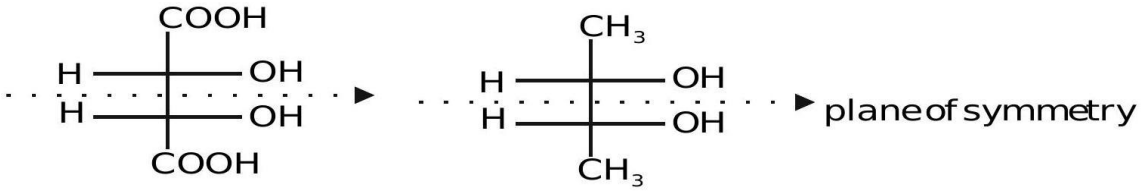


17.



When least priority group present on horizontal line then Anti-clockwise direction, configaration is given as ' R '

19. Non-super imposable nor -mirror images are called 'diastereomers'
20. First compound possess chiral carbon 2 nd 3rd compound in options possess chiral axis
21. Due to non- super imposable and non- mirror image above pair is diastereomers.
22. Due to presence of plane of symmetry above compound is meso
23. because of presence of plane of symmetry in 1,2 compounds they are meso compounds



8. General Organic Chemistry Part – II

WORKSHEET – 1

Level – I

1) B 2) A 3) B 4) B 5) D 6) C 7) A 8) B
9) C 10) C

Level – II

1) C 2) D 3) D 4) C 5) B 6) D 7) A 8) D
9) B 10) D 11) B 12) A 13) C 14) D 15) B 16) D
17) D 18) B 19) A,B,C,D 20) A 21) B 22) A 23) A,D
24) A,C,D 25) a-p; b-q; c-s; d-r 26) A,C,D 27)a-q; b-r; c-s; d-p 28) 5

WORKSHEET – 2

Level – I

1) D 2) A 3) A 4) B 5) A 6) D 7) B 8) A
9)A 10) B

Level – II

1) A 2) B 3) A 4) A 5) D 6) D 7) B 8) C
9) D 10) C 11) A 12) C 13) D 14) C 15) B 16) A
17) B 18) A,C 19) A 20) A 21) A 22) a-s; b-s; c-p; d-q
23) D 24) B 25) B 26) 8 27) A,B,D 28) 7

WORKSHEET – 3

Level – I

1) A 2) D 3) A 4) D 5) D 6) C 7) B 8) D
9) A 10) C

Level – II

1) C 2) D 3) B 4) A 5) A 6) A 7) A 8) B
9) A 10) D 11) D 12) A 13) C 14) A 15) D 16) B
17) C 18) C,D 19)C ,D 20) a-q; b-p; c-p,q,s; d-r,s 21) D 22) D
23) B 24) C 25) A,B,C 26) A,B,D 27) A 28) D

WORKSHEET – 4

Level – I

1) 2 2) 1 3) 1 4) 2 5) 2 6) 3 7) 4 8) 3
9)2 10) 4

Level – II

1) D 2) B 3) D 4) D 5) A,D 6) C 7) D 8) D
9) A 10) D 11) D 12) A 13) C 14) C 15) A 16) C
17) C 18) A,B,C,D 19) D 20)C 21) A 22)a-q; b-p; c-s; d-r
23) A,B,D 24) 8 25) 1,2,4 26) D 27) C 28) D